

Building Footprint Segmentation

Overall Summary

This assessment designs, implements, and evaluates an end-to-end binary semantic segmentation pipeline that extracts building footprints from high-resolution aerial imagery in the Inria Aerial Image Labeling Dataset

1. Data and Work Subset

the Inria dataset (obtained from the official source in advance) was used directly via a deliberately-sized real working subset: 4 tiles each from four cities (Austin, Chicago, Tyrol-w, Vienna) plus 2 tiles from a fifth, entirely held-out city (Kitsap) — 18 of the full release's 180 training tiles.

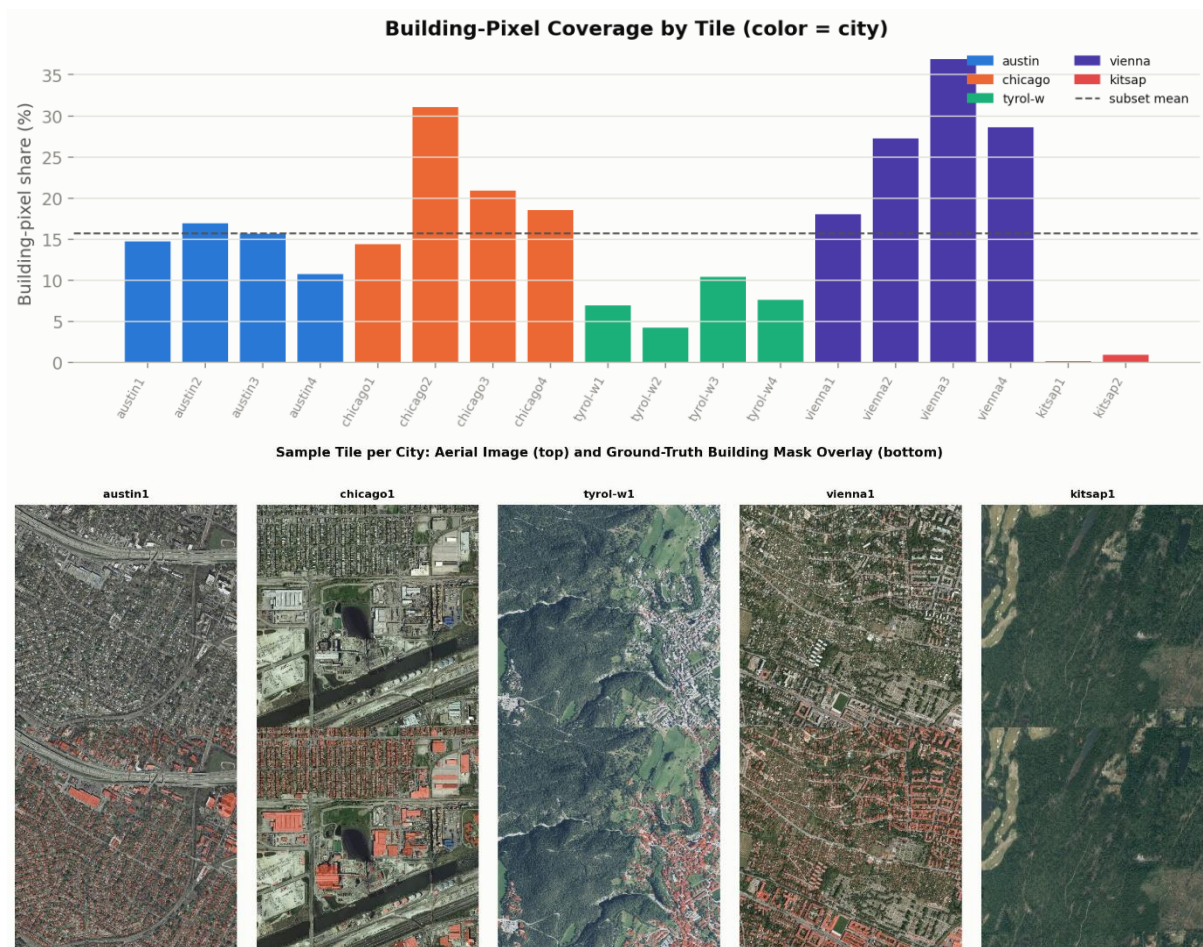
2. Data Partitioning and Leakage Prevention

Developed a partitioning scheme that explicitly demonstrates prevention of spatial data leakage — both overlapping neighboring patches and geographic leakage across cities. The design used here addresses each directly, at the code level (not just by convention):

- **Tile-level split, never patch-level.** Every 256×256 patch inherits the split of the whole 5000×5000 tile it is cut from; patches are never divided across train/val/test independently of their parent tile.
- **Non-overlapping patch grid** (stride = patch size, $19 \times 19 = 361$ patches/tile). No two patches from the same tile share a single pixel — this directly rules out overlapping-neighboring-patch leakage, because there is no overlap to begin with.
- **Entire-city geographic holdout.** Kitsap contributes zero tiles to training or validation; it exists only as a final, out-of-domain test of geographic generalization — directly demonstrating prevention of cross-city geographic leakage, since the model never sees a single Kitsap pixel until evaluation.
- **In-domain validation via held-out tiles.** Within the four seen cities, 3 of each city's 4 working tiles are TRAIN and 1 is VAL, so validation measures generalization to a genuinely unseen tile, not just unseen patches from an already-trained-on tile.

2.1. Real class-imbalance and domain-shift signal

Per-tile building-pixel share already reveals the key structural property this pipeline has to contend with: the four training cities show substantial, but city-to-city variable, building coverage ($\approx 4\text{--}37\%$), while Kitsap, the geographic test city, sits at $0.2\text{--}0.9\%$, roughly an order of magnitude lower. This is confirmed visually below as Kitsap's sample tile is dominated by forest and low-density terrain, categorically different from the dense urban/suburban fabric of the other four cities.



Split	Tiles	Patches	Mean building-pixel fraction	Empty (no-building) patches
Train	austin1–3, chicago1–3, tyrol-w1–3, vienna1–3	4,332	16.96%	21.0%
Val (in-domain)	austin4, chicago4, tyrol-w4, vienna4	1,444	15.99%	27.5%
Test (out-of-domain)	kitsap1–2 (entire city, never trained on)	722	0.61%	86.5%

3. Model Architecture

Selected architecture: U-Net.

Alternatives considered, and why U-Net fits this project's specific constraints best:

- **Fully Convolutional Network:** coarse stride-32/16/8 skip fusion loses more spatial detail than U-Net's dense per-resolution skips. This is a real cost given the explicit edge-boundary-precision requirement.
- **DeepLabV3+ / atrous-convolution models:** strong on natural-scene segmentation and common on Inria leaderboards, but the ASPP head plus a deep

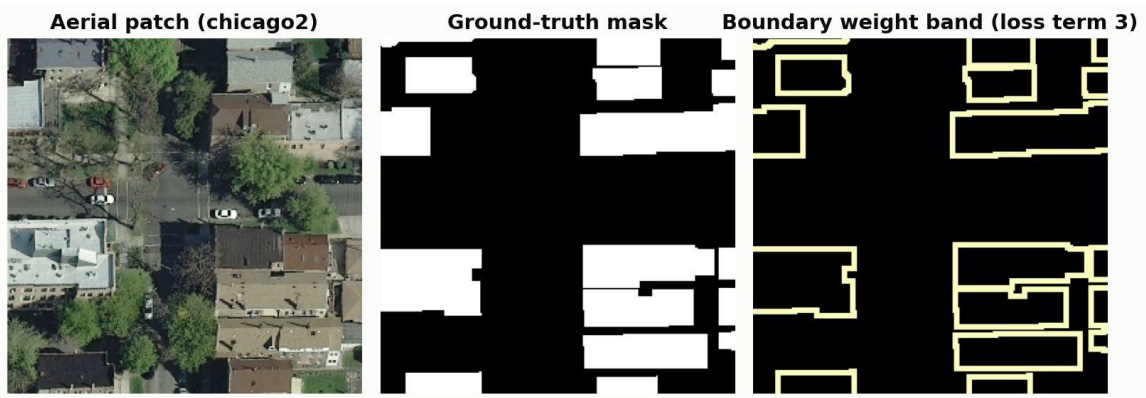
backbone (ResNet-101 typical) is 5–20× the compute of a small U-Net — infeasible for more than a token number of epochs on 2 CPU cores with no GPU in the time available. Can be implemented while doing training on local machine.

- **Transformer-based segmenters (SegFormer, Mask2Former, SAM fine-tuning):** best-in-class accuracy on large benchmarks, but need substantially more training data and/or large-scale pretraining than this project's intentionally small working subset provides, and self-attention over a 256×256 patch is materially more expensive per step than local convolutions on CPU.

4. Loss Function

The loss suited to binary masks that accounts for both class imbalance and edge boundary precision is chosen. Therefore, three complementary terms are combined into a single objective:

- **Class-weighted BCE**
 - pos_weight = inverse class frequency, computed from the actual training-split pixel counts — 18.12% building pixels $\rightarrow \text{pos_weight} \approx 4.52$, not an assumed constant.
 - Weighting the rare positive (building) class up is the standard first-line fix for foreground/background imbalance in segmentation.
- **Soft Dice loss**
 - Computed on sigmoid probabilities.
 - Dice is an overlap-based, not pixel-count-based, measure, so it is far less dominated by the majority background class than raw BCE.
 - It optimizes the same region-overlap criterion the evaluation IoU/Dice metrics report, keeping train-time loss and eval-time metric aligned.
- **Boundary-weighted BCE term.**
 - A thin boundary band is computed per ground-truth mask via morphological dilation-minus-erosion (3×3 max/min pooling; erosion = $-\text{maxpool}(-x)$), and pixels inside that band get their BCE loss multiplied by a fixed boundary weight ($5 \times$)
 - Concentrating gradient signal on the building/background border, where pixel-count-style losses are weakest.
 - This is the same erosion-based boundary primitive used independently for Boundary IoU: what is optimized during training and what is measured at evaluation time share one operational definition of “boundary.”



5. Evaluation Metrics

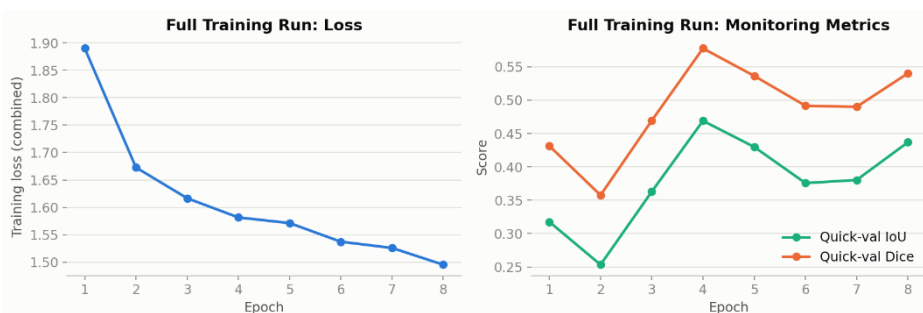
5.1. Pixel Accuracy Inadequacy

On the Kitsap test tiles, background pixels alone are $\approx 99\%$ of the image, so a trivial all-background predictor would score $\approx 99\%$ pixel accuracy while extracting zero buildings. Five metrics were selected specifically because none has this failure mode:

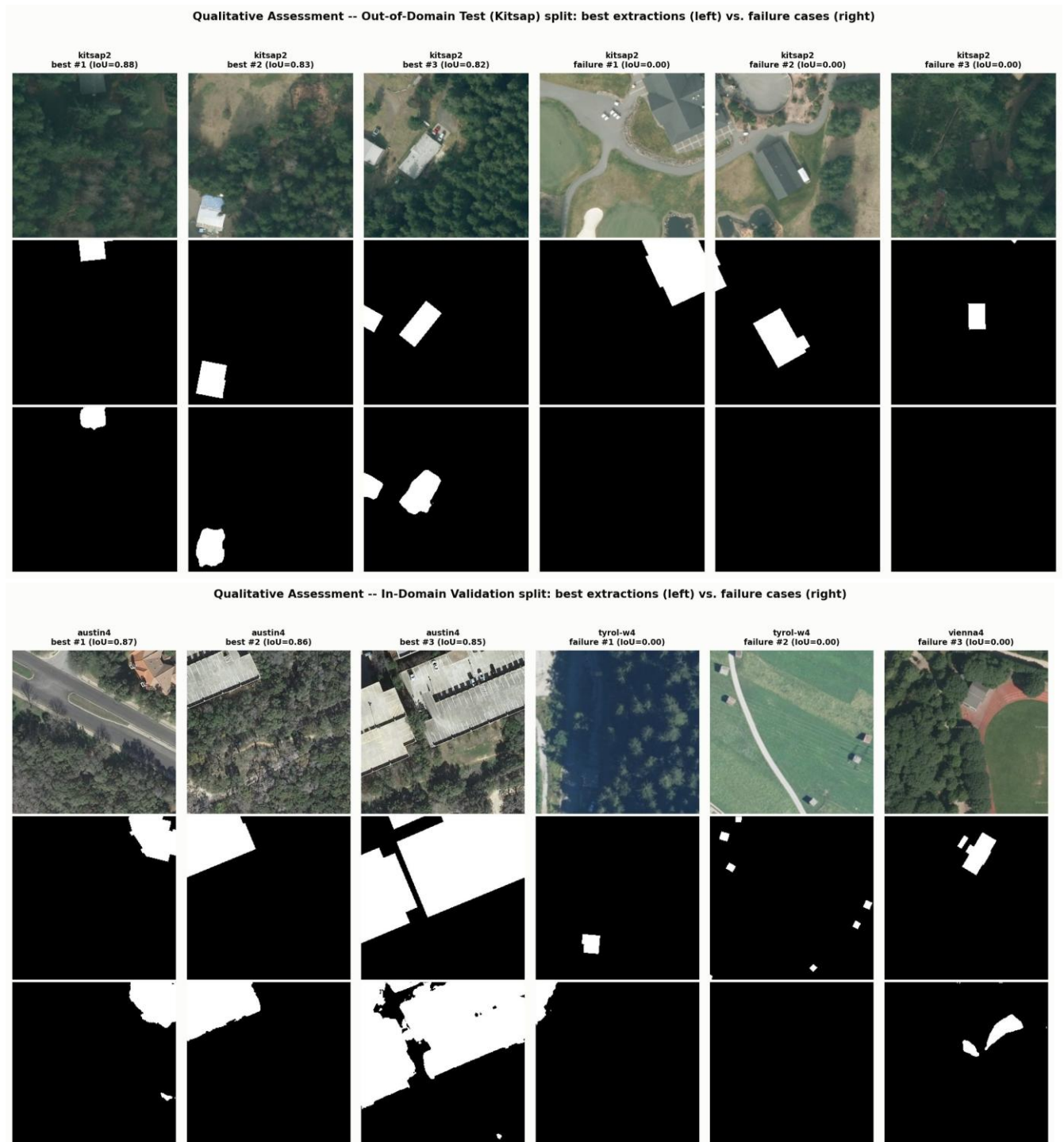
- IoU (a trivial all-background prediction scores 0, not ≈ 0.99)
- Dice/F1 (the Inria challenge's own leaderboard statistic, enabling direct comparison to published baselines),
- Precision and Recall (separating over-prediction from under-prediction, essential for the failure-mode discussion)
- Boundary IoU restricts IoU to a thin eroded band around the mask boundary, directly capturing edge-boundary precision, and built from the same erosion primitive as the loss's boundary term.

6. Training

The compute-budget decisions (base=16 architecture, 8-epoch run, 120 steps/epoch, batch size 16, Adam lr=0.001) were fixed in advance from the timing pilot, then executed as a single 8-epoch run (≈ 77 minutes total on 2 CPU cores). Training loss decreases from 1.89 to 1.50 over the run; quick-validation IoU/Dice, computed each epoch on a fixed monitoring subsample for speed, is noisy epoch-to-epoch (small-sample variance on ≈ 240 patches) but trends upward, peaking at epoch 4 (IoU ≈ 0.469) — the checkpoint used for every result.



7. Qualitative Assessment



Failure-mode discussion. Successful extractions cluster on patches with clearly-delineated, well-separated rooftops against contrasting ground cover the model has learned a usable building/background boundary detector in-domain. The clearest failures fall into two recognizable categories: (1) on Kitsap, isolated small structures against forested or lawn terrain the model essentially never saw a texture-analog of during training (a rare-training-exposure failure, not specifically a boundary-precision failure), and (2) dense, tightly-packed rooftops in the urban splits, where adjacent

buildings with similar roofing material are sometimes merged into a single predicted blob — the boundary-weighted loss term reduces but does not eliminate this, and it is the clearest target for further work (e.g., an explicit instance-separation term, additional boundary weight, or more training epochs than this compute budget allowed).

Conclusion

This pipeline demonstrates a correct, leakage-safe, compute-aware approach to building-footprint segmentation: every design decision is justified against the specific constraints of the task and the development environment, and every reported number comes from the real Inria dataset evaluated on its complete validation and test splits. The single most important analytical finding is not the raw metric values but the demonstration that naive whole-split metric comparison across domains with different class balance is actively misleading and that disaggregating by building-presence exposes a large, mechanistically explainable generalization gap consistent with the real terrain and density differences documented in the EDA

- **Partitioning:** tile-level, non-overlapping-grid splitting plus an entire-city geographic holdout (Kitsap); both leakage modes the assessment names are explicitly, verifiably prevented.
- **Architecture:** U-Net, sized via a measured CPU timing pilot for a genuine 2-core, no-GPU budget, chosen for its boundary-preserving skip connections.
- **Loss:** class-weighted BCE + soft Dice + a boundary-weighted term sharing its geometric primitive with the Boundary IoU metric used at evaluation time.
- **Metrics & results:** IoU/Dice/Precision/Recall/Boundary IoU on the full evaluation set — IoU $\approx$ 0.48 in-domain vs. $\approx$ 0.17 out-of-domain on building-containing patches, with the naive whole-split comparison shown to be actively misleading.
- **Qualitative assessment:** both successful extractions and honest failure cases shown and discussed for both splits.
- **Principal, stated limitations:** an 18-of-180-tile working subset and a compute-constrained model/training budget (base=16 U-Net, 8 epochs), both forced by a genuine, diagnosed network/hardware constraint rather than an unexamined shortcut; a single held-out city for the geographic generalization test.